

JEONGWOO SHIN

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RESEARCH INTEREST

Generative Modeling, Diffusion & Flow Matching, Reward-Finetuning of Diffusion Models, Stochastic Optimal Control, Diffusion/Flow Model Distillation, Long Video Generation, 3D-Consistent Video Generation.

EDUCATION

Graduate School of Data Science, Seoul National University (SNU)

Ph.D. Data Science

2024.03 – Current

- Advisor: Joonseok Lee
- Research: Generative Modeling, Diffusion & Flow Matching
- GPA: 3.70 / 4.30

Graduate School of Data Science, Seoul National University (SNU)

M.S. Data Science

2022.03 – 2024.02

- Thesis: *Self-Guided Masked Autoencoder*
- Advisor: Joonseok Lee
- GPA: 3.98 / 4.30

Seoul National University (SNU)

B.S. Landscape Architecture and Rural System Engineering

2016.03 – 2022.02

- Thesis: *Adsorptive Removal of Methylene Blue from Aqueous Solution Using Carbon Nanospheres*
- GPA: 3.59 / 4.30

PUBLICATIONS

* Equal contribution. † Corresponding author.

[C4] Efficient Generative Modeling beyond Memoryless Diffusion via Adjoint Schrödinger Bridge Matching

[Jeongwoo Shin](#), Jinhwan Sul, Joonseok Lee[†], Jaewoong Choi[†], Jaemoo Choi[†]

International Conference on Machine Learning (ICML) 2026

[C3] Equivariant Latent Alignment via Flow Matching under Group Symmetries

Sunghyun Kim*, Jaehoon Hahm*, [Jeongwoo Shin](#), Joonseok Lee[†]

International Conference on Machine Learning (ICML) 2026

[C2] Latent Diffusion Models with Masked AutoEncoders

Junho Lee*, [Jeongwoo Shin](#)*, Hyungwook Choi, Joonseok Lee[†]

International Conference on Computer Vision (ICCV) 2025

[C1] Self-Guided Masked Autoencoder

[Jeongwoo Shin](#), Inseo Lee, Junho Lee, Joonseok Lee[†]

Conference on Neural Information Processing Systems (NeurIPS) 2024

[P4] Efficient Adjoint Matching for Fine-tuning Diffusion Models

[Jeongwoo Shin](#)*, Dongsoo Shin*, Yuchen Zhu, Wei Guo, Yongxin Chen, Joonseok Lee[†], Jaewoong Choi[†], Jaemoo Choi[†]

Under Review at NeurIPS 2026

[P3] Diverse Motion Customization via Control-based Dynamic Optimization

Youngyoon Choi*, Kihyun Kim*, [Jeongwoo Shin](#)[†], Joonseok Lee[†]
Under Review at NeurIPS 2026

[P2] No More Hit-and-Hope: Optimal Initial Noise Forging via Noise Shuffling

Suho Ryu, [Jeongwoo Shin](#), Joonseok Lee[†]
Under Review at NeurIPS 2026

[P1] Optimal Crystal Flow for Fast and Reliable Crystal Generation

Seunghoon Yi, Youngwoo Cho, Jaehoon Hahm, [Jeongwoo Shin](#), Soo Kyung Kim[†],
Jaegul Choo[†], Joonseok Lee[†], Hongkee Yoon[†]
Under Review at NeurIPS 2026

AWARDS & SCHOLARSHIPS

[Research Fellowship for Ph.D. Candidates](#)

2025.09 – 2026.08

Principal Investigator (PI) National Research Foundation of Korea (NRF)

- Project: *Optimal Latent Space for Effective Image Generation*

- Competitive individual research grant for Ph.D. students, funded by the Ministry of Education.

- Research on optimal latent spaces for image generation.

- Proposed a stochastic optimal control based generative process, resulting in [P4].

RESEARCH EXPERIENCE

[Data Refinement using Multi-Modal Foundation Models](#)

2023.11 – 2024.09

Samsung Electronics Co., Ltd.

- Collaborative research project with Samsung Electronics.

- Worked on data refinement with large multi-modal foundation models for personalized search and generation.

TEACHING EXPERIENCE

TA, SNU

M3239.005300: Machine Learning and Deep Learning I

2023 Spring

LANGUAGES

Programming: Python, C, C++

Deep Learning: PyTorch, TensorFlow, NumPy

Natural Languages: English (Fluent), Korean (Native)

REFERENCES

Joonseok Lee

M.S. & Ph.D. Advisor (SNU)

2022.03 - Current

joonseok@snu.ac.kr